Progress toward a vaccine for extraintestinal pathogenic E. coli (ExPEC) II: efficacy of a toxin-autotransporter dual antigen approach.

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Extraintestinal pathogenic Escherichia coli (ExPEC) is a leading cause of worldwide morbidity and mortality, the top cause of antimicrobial-resistant (AMR) infections, and the most frequent cause of life-threatening sepsis and urinary tract infections (UTI) in adults. The development of an effective and universal vaccine is complicated by this pathogen's pan-genome, its ability to mix and match virulence factors and AMR genes via horizontal gene transfer, an inability to decipher commensal from pathogens, and its intimate association and co-evolution with mammals. Using a pan virulome analysis of >20,000 sequenced E. coli strains, we identified the secreted cytolysin α-hemolysin (HlyA) as a high priority target for vaccine exploration studies. We demonstrate that a catalytically inactive pure form of HlyA, expressed in an autologous host using its own secretion system, is highly immunogenic in a murine host, protects against several forms of ExPEC infection (including lethal bacteremia), and significantly lowers bacterial burdens in multiple organ systems. Interestingly, the combination of a previously reported autotransporter (SinH) with HlyA was notably effective, inducing near complete protection against lethal challenge, including commonly used infection strains ST73 (CFT073) and ST95 (UTI89), as well as a mixture of 10 of the most highly virulent sequence types and strains from our clinical collection. Both HlyA and HlyA-SinH combinations also afforded some protection against UTI89 colonization in a murine UTI model. These findings suggest recombinant, inactive hemolysin and/or its combination with SinH warrant investigation in the development of an E. coli vaccine against invasive disease.

Distribution and antibiotic resistance profile of key Gram-negative bacteria that cause community-onset urinary tract infections in the Russian Federation: RESOURCE multicentre surveillance 2017 study.

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Urinary tract infections (UTIs) are among the most common bacterial infections in clinical practice. This RESOURCE (pathogen distribution and antibiotic RESistance prOfile of key Gram-negative bacteria caUsing community-onsEt URinary traCt) study aimed to determine the antimicrobial resistance profile of Gram-negative bacteria isolated from outpatient urine samples collected across the Russian Federation. A total of 96 781 urine samples were collected from 520 cities in the Russian Federation between 01 January 1 and 31 December 2017. Antibiotic susceptibility was performed using semi-automated analysers. The mean age of the study population was 40.9 years; 80.2% were female and 19.8% were male. Of the uropathogens that were isolated, 64.2% were Gram-negative bacteria. Among these, Escherichia coli (E. coli) was the most common (49.1%) followed by Klebsiella pneumoniae (9.5%), Proteus mirabilis (2.9%), Pseudomonas aeruginosa (1.7%), and Enterobacter spp. (1.0%). Of the antibiotics that were tested, 50% of the isolated E. coli strains were resistant to ampicillin, 30.3% to co-trimoxazole, 26.2% to aztreonam, 28.8% to levofloxacin, and 21% to cefuroxime. Conversely, E. coli was highly susceptible to imipenem (0.7% resistant strains isolated), amikacin (0.9%), nitrofurantoin (4.5%), and fosfomycin (1.2%). The most active antimicrobials against Klebsiella pneumoniae were imipenem (6.8% resistant strains) and colistin (0.5%), while piperacillin/tazobactam (4.2%), cefoperazone/sulbactam (3.1%) and imipenem (0%) were the most active agents against Proteus mirabilis. The antimicrobials showing the highest activity against Pseudomonas aeruginosa were colistin (10.7% resistant strains) and aztreonam (0%), while piperacillin/tazobactam (7.1%) and

cefoperazone/sulbactam (2.3%) showed the highest activity against Enterobacter spp. The prevalence of fluoroquinolone and cephalosporin resistance among common UTI-causing Gram-negative bacteria highlights the growing challenge of successfully treating community-onset UTIs.

[Prevalence and bacterial susceptibility of community acquired urinary tract infection in University Hospital of Brasília, 2001 to 2005].

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Urinary tract infection is among the most common infectious diseases in clinical medicine, and knowledge of its epidemiology and the sensitivity profile of the etiological agents is mandatory. The aim of this study was to identify the most frequent etiological agents and the profile of sensitivity to antimicrobial agents of the bacteria isolated from urine cultures from outpatients at the University Hospital of Brasília between 2001 and 2005. From analyses at the hospitals microbiology laboratory, there were 2,433 positive urine cultures. Escherichia coli was the most commonly isolated bacteria (62.4%), followed by Klebsiella pneumoniae (6.8%) and Proteus mirabilis (4.7%). Escherichia coli showed the highest sensitivity to amikacin (98.6%), gentamicin (96.2%), nitrofurantoin (96.3%) and the quinolones ciprofloxacin (90.9%) and norfloxacin (89.8%), with low sensitivity to sulfamethoxazole-trimethoprim (50.6%). The others bacteria presented similar sensitivity profiles. In conclusion, Escherichia coli was the most commonly isolated bacteria, and it was highly sensitive to aminoglycosides, nitrofurantoin and quinolones.

Evolving Enterococcus faecalis Biofilms and Urinary Tract Infection Relapse: Does Vaginal Estrogen Matter?

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Enterococcus faecalis urinary tract infection (UTI) is common in postmenopausal females and these bacteria create biofilms that may reduce treatment efficacy. The role of local vaginal estrogen therapy in susceptibility to E. faecalis infection is unclear. The aim of the study was to evaluate differences in the treatment of relapsing E. faecalis UTIs in postmenopausal women using vaginal estrogen compared to premenopausal women. This was a secondary analysis of a retrospective cohort study of 71 female ambulatory patients seen within the gynecology or urogynecology practices between 2011 and 2020. Patients included had symptomatic E. faecalis UTI and a diagnosis of recurrent UTI. Patients with asymptomatic bacteriuria and concurrent pregnancy were excluded. Data was retrieved by chart review, stored, and analyzed utilizing descriptive statistics. A 2-sided Fisher exact test was performed to compare outcomes between postmenopausal and premenopausal patients and the prescription of additional rounds of antibiotics for relapse. Within this cohort, 57.8% were postmenopausal and 42.2% were premenopausal. There was no statistically significant difference in the need for additional antibiotics between postmenopausal and premenopausal patients (10.8% vs 14.3%, $P = 0.72$), postmenopausal patients not using vaginal estrogen and premenopausal patients (0% vs 14.3%, $P = 0.28$), postmenopausal patients using vaginal estrogen and premenopausal patients (20% vs 14.3%, $P = 0.70$), and among postmenopausal vaginal estrogen users and nonusers (20% vs 0%, $P = 0.11$). A small percentage of premenopausal and postmenopausal patients with recurrent UTI required additional antibiotics for E. faecalis relapse. However, there are no statistically significant differences between our estrogen-deficient or estrogenized postmenopausal patients, and premenopausal patients.

Continuing the search for bacterial urovirulence factors.

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Bacteria that commonly cause infections of the normal urinary tract (eg Escherichia coli, Proteus mirabilis and Staphylococcus saprophyticus) do so because they possess specific urovirulence factors. Adhesions of various types (often fimbriae) seem to be the most important of these. In E. coli several other factors have been recognized, and sub-sets of defined uropathogenic clones exist. On the other hand, urovirulence determinants are less easy to distinguish in species such as S. epidermidis and Klebsiella pneumoniae, that rarely cause such infections, or are pathogenic only in the presence of some abnormality or deficiency in host defences.

Complete Genome Sequence of the Uropathogenic Escherichia coli Strain NU14.

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Escherichia coli is the most common bacterium causing urinary tract infections in humans. We report here the complete genome sequence of the uropathogenic Escherichia coli strain NU14, a clinical pyelonephritis isolate used for studying pathogenesis.

Urinary tract infection: an overview.

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Urinary tract infection (UTI) remains very common. As many as 50% of women report having had at least one UTI in their lifetimes. Urinary tract infection is the most common cause of infection in nursing home residents and the most common source of bacteremia in the elderly population. Urinary tract infection occurs in patients with structurally or functionally abnormal urinary tracts (complicated UTI) and in patients with anatomically normal urinary tracts (uncomplicated UTI). Escherichia coli (E coli) is the most common cause of uncomplicated UTI, whereas antibiotic-resistant Enterobacteriaceae, enterococci, and Candida species often are the causes of complicated UTI. In this article we review current concepts of the epidemiology, microbiology, pathophysiology, clinical manifestations, diagnosis, and treatment of urinary tract infection.

Reaching the End of the Line: Urinary Tract Infections.

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Urinary tract infections (UTIs) cause a substantial health care burden. UTIs (i) are most often caused by uropathogenic Escherichia coli (UPEC), (ii) primarily affect otherwise healthy females (50% of women will have a UTI), (iii) are associated with significant morbidity and economic impact, (iv) can become chronic, and (v) are highly recurrent. A history of UTI is a significant risk factor for a recurrent UTI (rUTI). In otherwise healthy women, an acute UTI leads to a 25 to 50% chance of rUTI within months of the initial infection. Interestingly, rUTIs are commonly caused by the same strain of E. coli that led to the initial infection, arguing that there exist host-associated reservoirs, like the gastrointestinal tract and underlying bladder tissue, that can seed rUTIs. Additionally, catheter-associated UTIs (CAUTI), caused by Enterococcus and Staphylococcus as well as UPEC, represent a major health care concern. The host's response of depositing fibrinogen at the site of infection has been found to be critical to establishing CAUTI. The Drug Resistance Index, an evaluation of antibiotic resistance, indicates that UTIs have become increasingly difficult to treat since the mid-2000s. Thus, UTIs are a "canary in the coal mine," warning of the possibility of a return to the preantibiotic era, where some common infections are untreatable with available antibiotics. Numerous alternative strategies for both the prevention and treatment of UTIs are being pursued, with a focus on the development of vaccines and small-molecule inhibitors targeting virulence factors, in the hopes of reducing the burden of urogenital tract infections in an antibiotic-sparing manner.

Tamm-Horsfall protein knockout mice are more prone to urinary tract infection: rapid communication.

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Human colon contains many bacteria that commonly colonize the perineum and frequently enter the urinary tract. Uropathogenic Escherichia coli are the most common cause of urinary tract infection. Type 1 fimbriated E. coli have been associated with cystitis, and P fimbriated E. coli with pyelonephritis. Factors involved in clearing bacteria from the urinary tract are poorly understood. Tamm-Horsfall protein (THP), the most abundant protein in mammalian urine, has been postulated to play a role in defense against urinary tract infection but definitive proof for this idea has been lacking. In this study, we generated THP gene knockout mice by the technique of homologous recombination, and examined if the THP-deficient (THP-/-) mice were more prone to urinary tract infection. Various strains of E. coli expressing type 1 or P fimbriae were introduced transurethrally into the bladders of the THP-/- and genetically similar wild-type (THP+/+) mice. Urine, bladder, and kidney tissues were obtained from the mice and cultured for bacterial growth. THP-/- mice inoculated with type 1 fimbriated E. coli had a longer duration of bacteriuria, and more intense colonization of the urinary bladder in comparison with THP+/+ mice. When inoculated with a P fimbriated strain of E. coli, the THP-/- mice showed no difference in kidney bacterial load when compared with the THP+/+ mice. These findings support the idea that THP serves as a soluble receptor for type 1 fimbriated E. coli and helps eliminate bacteria from the urinary tract.

Rapid Growth of Uropathogenic Escherichia coli during Human Urinary Tract Infection.

PMID: 29511075 | Indexed in PubMed

Uropathogenic Escherichia coli (UPEC) strains cause most uncomplicated urinary tract infections (UTIs). These strains are a subgroup of extraintestinal pathogenic E. coli (ExPEC) strains that infect extraintestinal sites, including urinary tract, meninges, bloodstream, lungs, and surgical sites. Here, we hypothesize that UPEC isolates adapt to and grow more rapidly within the urinary tract than other E. coli isolates and survive in that niche. To date, there has not been a reliable method available to measure their growth rate in vivo Here we used two methods: segregation of nonreplicating plasmid pGTR902, and peak-to-trough ratio (PTR), a sequencing-based method that enumerates bacterial chromosomal replication forks present during cell division. In the murine model of UTI, UPEC strain growth was robust in vivo, matching or exceeding in vitro growth rates and only slowing after reaching high CFU counts at 24 and 30 h postinoculation (hpi). In contrast, asymptomatic bacteriuria (ABU) strains tended to maintain high growth rates in vivo at 6, 24, and 30 hpi, and population densities did not increase, suggesting that host responses or elimination limited population growth. Fecal strains displayed moderate growth rates at 6 hpi but did not survive to later times. By PTR, E. coli in urine of human patients with UTIs displayed extraordinarily rapid growth during active infection, with a mean doubling time of 22.4 min. Thus, in addition to traditional virulence determinants, including adhesins, toxins, iron acquisition, and motility, very high growth rates in vivo and resistance to the innate immune response appear to be critical phenotypes of UPEC strains.IMPORTANCE Uropathogenic Escherichia coli (UPEC) strains cause most urinary tract infections in otherwise healthy women. While we understand numerous virulence factors are utilized by E. coli to colonize and persist within the urinary tract, these properties are inconsequential unless bacteria can divide rapidly and survive the host immune response. To determine the contribution of growth rate to successful colonization and persistence, we employed two methods: one involving the segregation of a nonreplicating plasmid in bacteria as they divide and the peak-to-trough ratio, a sequencing-based method that enumerates chromosomal replication forks present during cell division. We found that UPEC strains divide extraordinarily rapidly during human UTIs. These techniques will be broadly applicable to measure in vivo growth rates of other bacterial pathogens during host colonization.